

When a Control Does Not Control: A Mechanism-Derived Refutation of a False Positive in p -adic Cryptanalysis of the ECDLP

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Abstract

Pipelines that fan out many structural probes against a hard problem and surface the most anomalous are increasingly common, and they are false-positive engines: with enough encodings and transforms an apparent $5\text{--}11\sigma$ effect is cheap. We give a worked example from a systematic negative exploration of p -adic lifting attacks on the elliptic curve discrete logarithm problem (ECDLP) for ordinary curves of the kind underlying ECDSA. A Gowers U^2 / Fourier analysis of the Teichmüller lift error $\delta(k)$ produced a persistent $5\text{--}11\sigma$ anomaly on the original sweep (and $\sim 3\text{--}14\sigma$ on a clean controlled set, with a standardized effect size ≈ 44 —against a 300-draw empirical null, not a Gaussian tail—on a confirmatory curve with $n \approx 10^4$) that a standard permutation control certified as “real, index-ordered structure.” We show this was a *control that cannot separate the hypotheses it was meant to test*. Two *mechanism-derived* nulls—one per candidate explanation—localize the entire signal to a single exact, public identity (the antisymmetry $\delta(k) + \delta(n - k) = [n]\tau(G)$): the signal vanishes when the reflection is removed, and—our key positive control—synthetic sequences carrying *only* that reflection reproduce the elevation and its growth with n (and a power-law Fourier decay of the same order, $\beta \approx 0.4$). We then give an inertness theorem, an explicit instance of a principle due to Silverman and to Gadiyar–Padma, that explains why every such anomaly on these curves must reduce to artifact-or-identity: for $\gcd(n, p) = 1$ the only secret-dependent part of any section lift error is $[k]c$, governed by $k \bmod p^{e-1}$, which is not determined by the public data (E, G, Q) over the natural lift (and, we note separately from the theorem, for which no efficient algorithm from (E, G, Q) is known). The single inequality $\gcd(n, p) = 1$ is sharp: when it fails ($n = p$, anomalous curves) it is the *ambiguity* that is annihilated— $[p]c = O$ for every kernel element c —so $[p]\tau(G)$ becomes a section-independent, secret-linear invariant and the same machinery recovers k with 100% success—Smart’s attack. The contribution is the methodology a results-free negative exploration must practice: a control is valid only against a named alternative hypothesis, nulls must be derived from mechanisms, and a structural reason for inertness beats an accumulation of passed tests.

1 Introduction

A recurring temptation in cryptanalysis is to lift a hard problem into a richer setting—a p -adic ring, a formal group, a higher-genus Jacobian—hoping that structure invisible over $\mathbb{F}_p$ becomes visible and exploitable. For the ECDLP this hope is well mapped: Silverman’s “four faces of lifting” [2, 1] catalogues the local/global $\times$ torsion/non-torsion strategies and the obstruction each meets, and

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the only p -adic local lift that ever broke an ECDLP is Smart’s attack on anomalous curves [4, 5, 6], where the group order equals the characteristic.

Automating such explorations sharpens an old hazard. A pipeline that computes many encodings of a lift invariant and reports the most unusual will, after enough nulls, return an “anomaly,” and the analyst must decide whether it is signal. We conducted a long, deliberately exhaustive negative exploration of the canonical-lift attack surface for ordinary curves with large prime-order subgroups—the ECDSA regime—and hit exactly this case: a higher-order Fourier probe returned a clear $5\text{--}11\sigma$ effect on the original sweep, a textbook control endorsed it, and only a more careful experiment overturned it. The overturning, and the reason the anomaly *had* to be spurious, are the substance of this paper.

Contributions.

- A worked, quantitative example of a *control that does not control*: a permutation test that appears to certify structure but provably cannot separate the competing explanations (§4).
- *Mechanism-derived controls*: enumerate the candidate generating mechanisms and build one null per mechanism, including a *positive* control in which a synthetic model of the surviving mechanism reproduces the statistic quantitatively (§5). This localizes the anomaly (a $\sim 3\text{--}14\sigma$ elevation on the clean controlled set) to one exact public identity.
- An inertness theorem (§7) explaining why such anomalies are forced: for $\gcd(n, p) = 1$ the secret-dependent part of any section lift error is $[k]c$, governed by $k \bmod p^{e-1}$, which the public data (E, G, Q) does not determine over the natural lift (and which no known efficient algorithm computes from it—a state-of-knowledge remark, kept separate from the theorem). We present it as the explicit instance it is of [2, 3], with proof and verification.
- A sharp-boundary demonstration (§8): the same construction on an anomalous curve ($\gcd(n, p) = p$) recovers the secret with 100% success. The theorem’s one hypothesis is exactly the line between inert and broken.

The theorem is not new mathematics; its role is to certify, after the fact, that the methodological suspicion was structurally warranted.

2 Background and notation

Let $p \geq 5$ be prime and $E/\mathbb{F}_p$ an elliptic curve with $\#E(\mathbb{F}_p) = N$. Fix $G \in E(\mathbb{F}_p)$ of order n with $\gcd(n, p) = 1$, and $Q = kG$; the ECDLP is to recover $k \in \mathbb{Z}/n\mathbb{Z}$ from (E, G, Q) . For $e \geq 1$, reduction modulo p on a fixed Weierstrass lift gives the exact sequence of abelian groups

$$0 \longrightarrow \widehat{E}(p\mathbb{Z}/p^e) \longrightarrow E(\mathbb{Z}/p^e) \xrightarrow{\text{red}} E(\mathbb{F}_p) \longrightarrow 0, \quad (1)$$

with kernel of reduction $\widehat{E}(p\mathbb{Z}/p^e)$ a p -group of order p^{e-1} [9]. The formal parameter $z = -X/Y$ identifies $\widehat{E}$ with $(p\mathbb{Z}/p^e, +_F)$ ($+_F$ the formal group law), and the formal logarithm $\log: \widehat{E} \rightarrow (p\mathbb{Z}/p^e, +)$ is an isomorphism with $\log([m]u) = m \log(u)$. For $E: y^2 = x^3 + b$ the invariant differential is $\omega = (1 + 3bz^6 + \dots) dz$, so $\log(z) = z + \frac{3b}{7}z^7 + \dots$; for $z \in p\mathbb{Z}$ and $e \leq 4$ the correction vanishes ($z^7 \in p^7\mathbb{Z}$), hence $\log(z) \equiv z \pmod{p^e}$ and kernel arithmetic is additive in z at our precision.

A *section* is a map $\tau: E(\mathbb{F}_p) \rightarrow E(\mathbb{Z}/p^e)$ with $\text{red} \circ \tau = \text{id}$; the Teichmüller and Hensel lifts are sections, generally not homomorphisms. We use the x -Teichmüller / Hensel section throughout;

it satisfies $\tau(-P) = -\tau(P)$ (the two Hensel roots in y are negatives), a property we invoke once below. The *lift error* (a 1-cochain valued in $\widehat{E}$ by (1)) is

$$\delta_\tau(k) := [k] \tau(G) - \tau(kG) \in \widehat{E}(p\mathbb{Z}/p^e). \quad (2)$$

We write $\delta(k)$ for the Teichmüller case and $z(\delta(k)) \in p\mathbb{Z}/p^e$ for its formal parameter; for an integer $z \in \mathbb{Z}/p^e$, $\text{digit}_j(z)$ is the coefficient of p^j in the base- p expansion (digit 0 is the units place, identically 0 here since $z \in p\mathbb{Z}$).

Prior art. That a p -adic lift transports the ECDLP without weakening it, that the canonical (Teichmüller) lift carries no information, and that “solving the DLP is equivalent to constructing a non-canonical lift” are due to Silverman [2, 1] for elliptic curves and to Gadiyar–Maini–Padma and Gadiyar–Padma [3] for $\mathbb{F}_p^\times$. Smart, Satoh–Araki and Semaev [4, 5, 6] give the anomalous attack; the canonical lift is Lubin–Serre–Tate [7], its computational instance Satoh’s point counting [8]. Our theorem repackages these in the lift-error language.

3 The exploration, in brief

The full study ran forty-seven numbered phases against the canonical-lift / formal-group attack surface for $j = 0$ (CM by $\mathbb{Z}[\omega]$) and generic ordinary curves; Appendix A indexes them. Probes included pseudorandomness of δ (DFT flatness, maximal Berlekamp–Massey complexity, uniform p -adic valuation, no polynomial/Mahler/multiplicative fit); a homomorphic section from H^2 vanishing (which exists but, being a homomorphism, annihilates the attack); an explicit Serre–Tate canonical lift validated by $\delta_{\text{can}} \equiv 0$; HNP/lattice (LLL, BKZ) attacks on the cocycle; MOV embedding degree; ordinary/supersingular bridges; genus-2 Jacobian covers; Weil pairings; Iwasawa μ ; and a Frobenius/unit-root analysis. Every one was inert. Two exact relations survived, both *public* (computable from the curve alone) and both giving no advantage: an automorphism-equivariance of the leading δ -digit, and the antisymmetry

$$\delta(k) + \delta(n - k) = [n] \tau(G), \quad (3)$$

which holds for any section with $\tau(-P) = -\tau(P)$ (so that $\tau(kG) + \tau(-kG) = O$), and which we confirmed holds *exactly* in the z -coordinate—e.g. for $p = 97$ ($n = 79$; a curve distinct from the $p = 67$, $n = 79$ entry of Table 1), $z(\delta(k)) + z(\delta(n - k))$ takes a single value across all 39 reflection pairs $(k, n - k)$.

4 A false positive: the Gowers anomaly

Encode digit j of the lift error as a phase, $f_j(k) = \exp(2\pi i \text{digit}_j(z(\delta(k)))/p)$ for $k = 1, \dots, n - 1$, and measure the Gowers U^2 norm [10] against an S^1 -uniform random baseline (definitions and counts in §6). Across many primes the U^2 norm was elevated $5\text{--}11\sigma$ on the original exploratory sweep (Phase 37, a mixed bag of curves later found to include supersingular and anomalous cases); on the clean controlled set of §5 the elevation is $\sim 3\text{--}14\sigma$ ($z(\text{real vs rand})$ ranges 3.3 to 13.6). U^3 and higher sat at baseline. The leading Fourier coefficient decayed as $|\hat{f}(\alpha)| \sim Cn^{-\beta}$ with $\beta < 1/2$, on CM, $j=1728$, and non-CM curves alike. An internal report concluded this was a genuine, previously unnoticed degree-1 structural property of the Teichmüller cocycle “not explained by any universal identity.”

The evidence offered was a permutation control: randomly permuting the *values* of f_j destroyed the elevation, read as proof that the structure lived in the index ordering, hence was “real.” This inference is invalid, and seeing why is the point of the paper.

5 Mechanism-derived controls

The principle. A control is a null model, and a null tests exactly one alternative: the one under which the statistic is recomputed. The value-permutation control recomputes U^2 under “the multiset of digit values, in random order,” so it separates “structure in the value distribution” from “structure in the ordering”—and nothing else. But at least three mechanisms predict ordered structure here, and the permutation kills all three at once, so it cannot attribute the signal to any one:

- (H1) *Prefix-sum artifact.* $\delta(k) = -\sum_{i < k} c(iG, G)$ is a running sum; summation is a low-pass filter ($\hat{s}(\xi) = \hat{d}(\xi)/(1 - e^{2\pi i\xi/n})$, a $1/\xi$ pole at $\xi = 0$), so *any* increments with this multiset, once summed, carry elevated low-frequency U^2 .
- (H2) *Antisymmetry shadow.* The exact reflection (3) couples Fourier mode α with $-\alpha$, a rigid constraint that elevates U^2 .
- (H3) *Genuine new structure,* the claimed explanation.

We build one null per mechanism, each derived from how that mechanism generates ordering. All numbers below are on the clean ordinary set of Table 1 (Phase 43), under the pre-registered protocol of §6.

H1 (increment shuffle). $s(k) = z(\delta(k))$ is the exact cumulative sum of its first differences $d(k) = s(k+1) - s(k)$. Shuffle d , re-sum in the new order, recompute U^2 ; the increment multiset is identical, only the order changes. The shuffled-increment prefix sums sit *at* the random baseline ($z(\text{incShuf vs rand}) \in [-0.3, +0.5]$ on every curve), while the true ordering sits $+3.5$ to $+12.5$ above them, growing with n (Table 1, cols. 4–5). H1 is refuted: the digit nonlinearity destroys the walk’s low-frequency mass, so the signal is in the *specific* order, not in summation.

H2 (half-sequence). The reflection (3) relates each k to $n - k$. On the first half $k = 1, \dots, \lfloor (n-1)/2 \rfloor$ every reflection partner lies outside the window, so H2 predicts no internal constraint there while H3 predicts the signal persists. Re-running the increment-shuffle control on the half collapses the elevation to noise ($z(\text{half vs incShuf}) \in [-1.3, +1.5]$, no growth in n ; Table 1, col. 6).

H2, positive control (the decisive test). Absence of signal on the half is necessary but not sufficient. We therefore test H2 *constructively*: generate synthetic sequences that impose *only* the reflection, $s(k)$ i.i.d. uniform on the first half and $s(n - k) = C - s(k)$ with $C = z([n]\tau(G))$, carrying no other structure, and measure their U^2 . If reflection alone is the mechanism, these must reproduce the real statistic. They do, quantitatively: the synthetic U^2 is elevated $+3.8$ to $+12.5$ over the random baseline—tracking the real elevation across the whole range of n —and the real sequences are statistically indistinguishable from them ($z(\text{real vs syn}) \in [-1.4, +1.5]$; Table 1, cols. 7–8). The match on the U^2 elevation and its growth with n is the substance of the positive control, and it is exact (cols. 7–8). The finer digit-dependent Fourier decay is only *partially* reproduced: fitting $|\hat{f}| \sim Cn^{-\beta}$ across the seven curves, the real sequences give $\beta_{\text{real}} \approx \{0.34, 0.45, 0.36\}$ for

digits 1, 2, 3, whereas the reflection-only synthetic draws give a nearly flat $\beta_{\text{syn}} \approx \{0.41, 0.41, 0.40\}$. Reflection alone thus reproduces the overall power-law decay ($\beta \approx 0.4$, the same order as the real exponents) but not its digit-to-digit variation; we do *not* claim the spectra match in detail. The residual digit-to-digit variation is, however, consistent with finite-size fit scatter rather than secondary structure: resampling the seven-curve fit on reflection-only sequences gives β_{syn} with mean ≈ 0.40 and standard deviation ≈ 0.06 *with no digit dependence built in*, and each real exponent $\{0.34, 0.45, 0.36\}$ lies within ≈ 1 standard deviation of that mean. Theorem 1 forecloses its being cryptanalytic in any case (a deterministic post-processing of the public digit map of a public object cannot leak the secret). The decisive evidence that H2 is the mechanism is the U^2 match, which is genuine. A post-hoc confirmatory run on a large-prime-order curve ($p = 10477$, $n = 10639$ prime; Phase 47, last row of Table 1) extends the pattern by more than an order of magnitude in n : the real elevation reaches +43–44, the reflection-only synthetics track it at +44.5–44.8, and real vs. synthetic stays within ± 2.1 .

The growth law is itself an artifact signature. The elevation does not merely persist with n ; it *grows*, from ~ 3 at $n = 79$ to ~ 44 at $n = 10639$. It is tempting to read growth-with- n as accumulating signal—this is precisely what the internal Phase-37 report did. The opposite is true: the growth is forced by the reflection alone and carries no curve information, as the following closed form shows.

Proposition 1 (The reflection inflates the U^2 mass by exactly $3/2$). *Let $f: \mathbb{Z}/L \rightarrow S^1$. For f with i.i.d. S^1 -uniform entries, $\mathbb{E}[\|f\|_{U^2}^4] = 2/L + o(1/L)$. For f subject to only the antisymmetry constraint $f(L+1-k) = \beta \overline{f(k)}$ ($\beta \in S^1$ fixed, the free half i.i.d. S^1 -uniform)—the reflection-only synthetic of the positive control— $\mathbb{E}[\|f\|_{U^2}^4] = 3/L + o(1/L)$. The ratio is exactly $3/2$, independent of L and of the value alphabet. Consequently the standardized synthetic-vs-random gap grows as $z(\text{syn}/\text{rand}) = \Theta(\sqrt{L})$.*

Proof sketch. $\|f\|_{U^2}^4 = \sum_{\xi} |\hat{f}(\xi)|^4$ with $\hat{f}(\xi) = \frac{1}{L} \sum_k f(k) e^{-2\pi i k \xi / L}$. For i.i.d. f the $L - 1$ nonzero modes are asymptotically independent circularly-symmetric complex Gaussians with $\mathbb{E}|\hat{f}(\xi)|^2 = 1/L$, so $\mathbb{E}|\hat{f}(\xi)|^4 = 2/L^2$ (two Wick pairings) and the sum is $2/L$. The constraint forces $\hat{f}(L-\xi) = \beta \overline{\hat{f}(\xi)}$, coupling each mode to its index-reversal partner; this added second-order structure raises $\mathbb{E}|\hat{f}(\xi)|^4$ from $2/L^2$ to $3/L^2$ (the coefficient 3 verified by direct second-moment measurement, Phase 48), and the sum to $3/L$. The mean U^2 ratio is then $(3/2)^{1/4}$; as the random-null standard deviation scales as $\Theta(L^{-3/4})$, the standardized gap scales as $\sqrt{L}$. $\square$

Numerically the constant is $z(\text{syn}/\text{rand}) \approx 0.43\sqrt{n}$ (verified p - and L -independent, $L \in [78, 10638]$; Phase 48), and $0.43\sqrt{n}$ reproduces the syn/rand column of Table 1 and the Phase-47 value (predicted 44.8 vs. observed 44.6 at $n = 10639$) within finite-size noise. So $\sqrt{n}$ growth is the *signature of the antisymmetry artifact*, not of an accumulating signal—the exact inference the original report inverted. The inflation is specific to the *index-reversing* coupling of (3): a constraint sending mode ξ to its reversal $-\xi$ raises the mass to $3/L$, whereas one sending it elsewhere (e.g. a fixed shift) leaves it at $2/L$ —a discriminator an auditor can apply directly.

Verdict. The anomaly is the Fourier-domain shadow of the exact, public identity (3): H2, not H1 or H3. It is real but not new, and—being a public consequence of the curve’s geometry that yields only a 2-to-1 search reduction—carries no cryptanalytic advantage. The varying “decay exponent” and its apparent curve-family dependence in the internal report are finite-size scatter around a single reflection-induced spectrum; nothing family-specific survives.

p	n	L	negative controls (z)			positive (z)	
			real/inc	inc/rand	half/inc	syn/rand	real/syn
67	79	78	+3.5..6.7	−0.3..0.2	−0.6..1.5	+3.8..4.1	−0.4..1.2
163	139	138	+3.7..7.4	+0.0..0.5	−0.4..1.0	+5.4..5.4	−0.9..1.5
211	199	198	+5.1..7.8	+0.1..0.3	−0.8..0.5	+5.8..5.9	−0.3..0.8
349	313	312	+5.4..7.1	−0.0..0.1	−1.3..0.2	+7.5..7.6	−1.4.. − 0.4
433	397	396	+8.2..9.6	−0.1..0.2	−0.4..0.6	+8.4..8.6	−0.1..0.7
577	613	612	+7.8..11.5	+0.1..0.2	−0.5..1.1	+10.2..10.4	−1.2..0.8
823	829	828	+10.5..12.5	+0.1..0.2	−0.9..0.0	+12.2..12.5	−0.9..0.7
10477 [†]	10639	10638	+42.7..44.1	−0.0..0.0	−1.1..1.8	+44.5..44.8	−2.0..0.7

Table 1: Phase 43, clean ordinary $j=0$ curves $y^2 = x^3 + 7$ ($p \equiv 1 \pmod{3}$, $n = \#E(\mathbb{F}_p)$ prime, so $\gcd(n, p) = 1$ and the curve is neither supersingular nor anomalous); L is the number of usable $k \in [1, n - 1]$ after the pre-registered degeneracy filter (§6); here $L = n - 1$ (no drops). Ranges are over digits $j = 1, 2, 3$. Negative controls: shuffling increments leaves U^2 at the random baseline (inc/rand) and the true ordering far above it (real/inc), which vanishes on the half (half/inc). Positive control: synthetic reflection-only sequences reproduce the elevation (syn/rand) and the real sequences match them (real/syn). The entire effect is the antisymmetry (3). The last row ([†], Phase 47, $p = 10477$) is a post-hoc confirmatory run under the identical protocol, reported separately from the pre-registered seven-curve set.

6 Statistical protocol

All quantities in §5 follow one pre-registered protocol (fixed before the clean re-run; seed 20260609). Precision $e = 4$; digits $j \in \{1, 2, 3\}$. The S^1 -random baseline draws 300 sequences $g(k) = e^{i\theta_k}$, θ_k uniform on $[0, 2\pi)$, of the same length L ; “ σ ” and “ z ” are the standard deviation and standardized deviation of the relevant null: incShuf uses 300 increment shuffles, syn uses 300 synthetic draws, rand uses the 300 baselines. The Gowers U^2 is computed exactly by its FFT formula. *Exclusion rule (pre-specified)*: a value $z(\delta(k))$ is dropped iff its projective lift is degenerate— Y a non-unit, or $z \not\equiv 0 \pmod{p}$ (a kernel element must have $z \equiv 0$); these are arithmetic failures of the pure-Python lift, flagged automatically, and on the clean ordinary set none occurred ($L = n - 1$ throughout). The clean set is the seven ordinary $j=0$ curves $y^2 = x^3 + 7$ with $p \equiv 1 \pmod{3}$ and $n = \#E(\mathbb{F}_p)$ prime (hence $\gcd(n, p) = 1$, neither supersingular nor anomalous); the smallest has $n = 79$, so $L \geq 78$ throughout. No further size-based filter is applied—every such curve in the run is reported. We make no multiplicity correction and therefore do *not* claim discovery from any single elevated U^2 ; the paper’s logic is the opposite—every elevation is traced to a named mechanism and shown to be identity-or-artifact, and the controls are designed to *remove* signal, not to find it. Phase 47 repeats this protocol verbatim on one large-prime-order curve ($p = 10477$, $n = \#E(\mathbb{F}_p) = 10639$ prime); being post-hoc, it is reported separately (marked [†] in Table 1). Throughout, z -values are standardized effect sizes against 300-draw empirical nulls, not calibrated Gaussian tail probabilities; we use them only comparatively, never as discovery statistics.

7 Why it had to be a false positive: an inertness theorem

For $\gcd(n, p) = 1$ the secret-dependent part of any section lift error is forced into a coordinate the attacker cannot compute, so any “structure” in δ that *is* computable from (E, G, Q) must reduce to a public identity or an artifact. We make this precise.

Lemma 1 (Canonical lift). *If $\gcd(n, p) = 1$, the order- n subgroup $\langle G \rangle$ lifts uniquely to $\langle \tilde{G} \rangle \subset E(\mathbb{Z}/p^e)$ of order n with $\text{red}(\tilde{G}) = G$, and $jG \mapsto [j]\tilde{G}$ is a homomorphism (so $\delta_{\tau_{\text{can}}} \equiv 0$). Explicitly $\tilde{G} = [m]\tau(G)$ for any section τ , where $m \equiv 1 \pmod{n}$, $m \equiv 0 \pmod{p^{e-1}}$.*

Proof. $E(\mathbb{Z}/p^e)[n]$ injects into $E(\mathbb{F}_p)[n]$ (an n -torsion element of $\hat{E}$ is killed by $\gcd(n, p^{e-1}) = 1$, hence trivial) and surjects onto it (étale n -torsion lifts by Hensel), so reduction is an isomorphism on n -torsion and $\tilde{G}$ is the unique order- n lift of G . Writing $\tau(G) = \tilde{G} + c$ with $c \in \hat{E}$, $[m]\tau(G) = [m \bmod n]\tilde{G} + [m \bmod p^{e-1}]c = \tilde{G}$. $\square$

Theorem 1 (Inertness). *Let τ be any section and $c := \tau(G) - \tilde{G} \in \hat{E}$. Then:*

- (a) $\delta_\tau(k) = [k]c - c_k$, where $c_k := \tau(kG) - [k]\tilde{G} \in \hat{E}$ depends only on the point kG .
- (b) $[k]c$ depends on k only through $k \bmod p^{e-1}$; $\log([k]c) = k \log(c)$ has additive order a power of p dividing p^{e-1} .
- (c) $\gcd(n, p^{e-1}) = 1$, so by CRT, for k uniform on $\mathbb{Z}/np^{e-1}\mathbb{Z}$ the residues $k \bmod n$ (fixed by Q) and $k \bmod p^{e-1}$ (governing $[k]c$) are independent. The public data (E, G, Q) fixes only $k \bmod n$; hence over this natural lift $[k]c$, a function of $k \bmod p^{e-1}$, is not determined by the public data.
- (d) δ_τ is a well-defined function of the instance (i.e. of $k \bmod n$) iff $c = 0$ (i.e. τ canonical on $\langle G \rangle$), in which case $\delta_\tau \equiv 0$. For $c \neq 0$, $\delta_\tau(k+n) - \delta_\tau(k) = [n]c = [n]\tau(G) \neq O$, the antisymmetry constant of (3).

Proof. (a) $[k](\tilde{G} + c) - ([k]\tilde{G} + c_k) = [k]c - c_k$. (b) $c \in \hat{E}$ has order dividing p^{e-1} ; apply $\log$, whose order as a map $k \mapsto k \log(c)$ on $(p\mathbb{Z}/p^e, +)$ is $p^e / \gcd(\log(c), p^e)$, a power of p dividing p^{e-1} since $\log(c) \in p\mathbb{Z}$. (c) Since $\gcd(n, p^{e-1}) = 1$, CRT gives $\mathbb{Z}/np^{e-1}\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/p^{e-1}\mathbb{Z}$, so for k uniform on the left the residues $k \bmod n$ and $k \bmod p^{e-1}$ are independent; as (E, G, Q) determines only $k \bmod n$, it leaves $k \bmod p^{e-1}$ (and hence $[k]c$) information-theoretically unconstrained over this lift. We make no claim of a reduction beyond this. (d) c_k is a function of kG , hence of $k \bmod n$; so δ_τ factors through $k \bmod n$ iff $k \mapsto [k]c$ does, iff $[n]c = O$, iff (as $\gcd(n, p^{e-1}) = 1$) $c = O$. The shift equals $[n]c = [n]\tau(G)$ since $[n]\tilde{G} = O$. $\square$

Corollary 1. *For $\gcd(n, p) = 1$ no section lift error yields an efficiently computable function of (E, G, Q) that depends on the secret: it is either identically zero (canonical) or not a function of the instance at all (non-canonical, ambiguity $[n]\tau(G) \neq O$). The p -adic canonical lift moves the ECDLP without weakening it.*

This is why the Gowers anomaly was forced to be artifact-or-identity. The only secret-bearing object, $[k]c$, is not publicly computable; the visible δ , taken as a function of the integer k , is either zero or periodic-with-shift $[n]\tau(G)$ —and that shift is the reflection (3) whose Fourier signature the probe rediscovered.

Remark (on clause (c)). We deliberately do *not* claim $[k]c$ is information-theoretically independent of (E, G, Q) : once a canonical representative $k \in [1, n-1]$ is fixed, $k \bmod p^{e-1}$ and hence $[k]c$ are determined. The claim is computational—no efficient algorithm produces $[k]c$ from the public data short of solving the ECDLP—which is what matters for security and what the experiments probe. Being a state-of-knowledge assertion, this claim is confined to the present remark rather than the theorem statement.

Regime	curves	$[n]$ on $\widehat{E}$	invariant $[n]\tau(G)$	ECDLP
Small ordinary, $\gcd(n, p) = 1$	$p = 31\text{--}163, j=0$	automorphism	$\neq O$, $\text{ord} = p^{e-1}$, public	inert
Anomalous (Smart), $n = p$	$y^2 = x^3 + 4x + 7/\mathbb{F}_{53}$	zero map	$\neq O$, section-indep., $\propto k$	broken, 12/12

Table 2: Phases 41–42. What vanishes at the boundary is the *ambiguity*, not the invariant. For $\gcd(n, p) = 1$, $[n]$ is an automorphism of the kernel: the well-definedness obstruction of Theorem 1(d) has maximal order and the only well-defined quantity is the public antisymmetry constant of (3). For $n = p$, $[n]$ annihilates the kernel: the section ambiguity $[p]c$ dies and $[p]\tau(G)$ becomes a well-defined, *nonzero*, secret-linear invariant—which Smart’s attack requires ($\varphi(G) \neq 0$). $\gcd(n, p) = 1$ is the exact boundary. Theorem 1 is regime-agnostic—it holds for any $\gcd(n, p) = 1$ —so these small ordinary curves (chosen for exact-arithmetic reach) suffice to verify it; the relevance to large-prime ECDSA curves is through the theorem’s generality, not through these particular orders.

8 The boundary is sharp

Theorem 1 hinges on $\gcd(n, p) = 1$, and what changes at the boundary is the action of $[n]$ on the kernel of reduction. Set $n = p$ (an anomalous curve, $\#E(\mathbb{F}_p) = p$; take $e = 2$): $[n] = [p]$ now *annihilates* the kernel— $[p]c = O$ for every $c \in \widehat{E}(p\mathbb{Z}/p^2)$, since $\log([p]c) = p \log(c) \in p^2\mathbb{Z}$ —whereas for $\gcd(n, p) = 1$ it acts as an automorphism. Two consequences follow. First, the *section ambiguity* dies: two sections differ pointwise by a kernel element, which $[p]$ kills, so $[p]\tau(P)$ is independent of the choice of section. Second, what survives is *not* zero: $\varphi(P) := \frac{1}{p} \log([p]\tau(P)) \bmod p$ is a homomorphism $E(\mathbb{F}_p) \rightarrow \mathbb{F}_p$, linear in the secret, and Smart’s attack is its evaluation, $k = \varphi(Q)/\varphi(G)$. The asymmetry with the ordinary case is exact: there the ambiguity has maximal order and the only well-defined quantity is the public constant of (3); here the ambiguity vanishes and the surviving invariant carries k . (Note that δ itself remains ill-defined as a function of $k \bmod p$: $\delta(k+p) - \delta(k) = [p]\tau(G) \neq O$. The object that becomes well-defined is $[p]\tau(P)$, not δ .) We verified both sides (Table 2): on every ordinary test curve the obstruction $[n]\tau(G)$ is a nonzero kernel element of *maximal* order p^{e-1} (valuation 1), while on the anomalous $y^2 = x^3 + 4x + 7$ over $\mathbb{F}_{53}$ (with $G = (0, 22)$) the value $z([p]\tau(G)) = 15 \cdot 53 \bmod 53^2$ is *section-independent* (identical under the Teichmüller- x , naive- x , and shifted- x sections) and nonzero—as it must be, since the attack divides by $\varphi(G) = 15$ —and Smart’s attack recovers k for 12/12 random secrets. The same code path that is inert for $\gcd(n, p) = 1$ is a total break the instant $n = p$.

The dichotomy is exhaustive, not merely sharp. One might ask whether an *intermediate* regime exists—a curve on which $[n]$ partially degenerates on the kernel, $0 \neq \ker([n]|_{\widehat{E}}) \subsetneq \widehat{E}$, giving a graded leak. On $E(\mathbb{F}_p)$ there is none. A partial kernel needs $1 \leq v_p(n) < e - 1$ with $v_p(n) \geq 1$, i.e. $p \mid n$; but $n \mid N = \#E(\mathbb{F}_p)$, so $p \mid n$ forces $p \mid N$, and by Hasse $N \in [(\sqrt{p} - 1)^2, (\sqrt{p} + 1)^2]$ the only multiple of p in range is $N = p$ itself for $p \geq 7$ (we verified no curve with $p \mid N \neq p$ exists for $p = 7, \dots, 23$; the lone exception $N = 2p$ occurs only at $p = 5$). Hence $v_p(n) \in \{0, 1\}$ and $[n]$ acts on $\widehat{E}$ as an automorphism ($p \nmid n$) or, on its p -part, the zero map ($n = p$), never partially. The boundary is a true dichotomy with no curves in between; a “graded inertness theorem” has empty interior here.

9 Numerical verification

Theorem 1 was checked on $y^2 = x^3 + 7$ at $e = 4$ for ten small ordinary curves over $\mathbb{F}_p$, $p = 31, \dots, 163$, all with $\gcd(n, p) = 1$ (Phase 41). These test orders are deliberately small and several are composite— $n \in \{21, 31, 79, 67, 79, 111, 129, 133, 86, 139\}$, five of them composite—chosen for exact-arithmetic reach rather than to model the large-prime ECDSA regime; since the theorem is regime-agnostic (it needs only $\gcd(n, p) = 1$), they suffice to verify it. (Here n denotes the order of the chosen G and may be a proper divisor of $N = \#E(\mathbb{F}_p)$, e.g. $n = 86$ with $N = 172$ at $p = 157$; the clean set of Table 1 instead requires $n = N$ prime.) Phase 47 additionally verifies every clause on one large-prime-order ordinary curve ($p = 10477$, $n = N = 10639$ prime): C_n again has valuation 1 and additive order exactly p^{e-1} , the antisymmetry is single-valued across all 5319 reflection pairs, and $L = n - 1$ with no degenerate drops. All kernel arithmetic is done through the additive z -coordinate to avoid projective degeneracy on deep kernel points. On every prime: the obstruction $C_n := z([n]\tau(G))$ is nonzero in $p\mathbb{Z}$ of valuation 1 and additive order exactly p^{e-1} (clauses (d),(b)); the antisymmetry $z(\delta(k)) + z(\delta(n - k)) \equiv C_n$ holds for all k (clause (d) / (3)); the integer-representative shift $z(\delta(k + n)) - z(\delta(k)) \equiv C_n$ holds on all eight arithmetically-reliable primes (a group identity; on two primes the deep multiply degenerates in exact arithmetic, flagged by a self-consistency probe); and $\gcd(n, p^{e-1}) = 1$ (clause (c)). The canonical case $\delta_{\text{can}} \equiv 0$ (Lemma 1) was confirmed in an exact p -adic computation. Anomalous recovery is Phase 42 (§8).

10 Discussion

For negative cryptanalysis. The reusable lesson is narrow and practical. (i) A control tests exactly one alternative—the one its null recomputes; before trusting it, enumerate the mechanisms that could generate the statistic and check whether the control separates them. The permutation test here failed because three mechanisms collapsed under it. (ii) Build one null per mechanism, derived from how that mechanism generates the statistic: H1 from the prefix-sum structure (shuffle increments, re-sum), H2 from the reflection’s reach (restrict to a half), and—decisively—a *positive* control synthesizing the surviving mechanism (reflection-only sequences) and showing it reproduces the statistic. (iii) Beware cumulative statistics: any running sum manufactures low-frequency autocorrelation, so “looks autocorrelated” is the null, not the finding. (iv) Where a structural theorem is available it bounds what controls can find; Theorem 1 says no honest control turns δ into an oracle for $\gcd(n, p) = 1$.

For automated and AI-assisted exploration. Pipelines that fan out many probes and surface the most anomalous are false-positive engines: with enough encodings and transforms an apparent $5\text{--}11\sigma$ effect is cheap, and a plausible-sounding control can launder it into a “finding.” The safeguard is not more probes but mechanism-derived nulls, a positive control for any surviving signal, and—where possible—a closed-form reason the probe must be inert. The theorem’s value here is exactly that: it turns “we tried forty things and they failed” into “here is the one-line reason they had to.”

Honest scope. Theorem 1 is not new mathematics. It is an explicit instance, in the lift-error language, of the principle of [2, 1, 3] that lifting transports the ECDLP; the anomalous degeneration is Smart’s attack [4]. It concerns section lift errors only; pairing-based functionals are a separate mechanism (inert here for the independent reason of huge embedding degree). We claim novelty only for the methodology—the mechanism-derived controls and the positive control that localizes the anomaly—and for the explicit identification of the obstruction with the antisymmetry constant.

11 Conclusion

A long negative exploration produced one seductive anomaly. A standard control endorsed it; mechanism-derived controls, capped by a positive control, killed it and attributed it entirely to an exact public identity; and an inertness theorem explains why no such anomaly on ordinary curves can be more than artifact-or-identity, with the boundary falling exactly at the anomalous curves where the lift attack genuinely works. The cryptanalytic verdict is the expected negative one. The transferable result is a discipline: name your alternative hypotheses, build a null for each from its mechanism, synthesize the survivor to confirm it, and prefer a structural reason for inertness over an accumulation of passed tests.

A Phase index

Phase(s)	Target	Verdict
1–2	injectivity / pseudorandomness of δ	injective, pseudorandom
3–4	Smart–SSSA; homomorphic section from H^2	section exists, attack does not
5–6	CM-linearization; HNP on cocycle	pseudorandom; no bias
7–10	MOV; poly fit; LLL coords; ord/ss bridge	inapplicable / no leak
11–13	genus-2 Jacobian; GLV-CRT; Weil pairing	no new lever
14	$\text{Aut}(E)$ -equivariance rank law of δ -digit	public, no advantage
15–20	triple cocycle; DFT; Berlekamp–Massey; valuation; mult.	all inert
21b	antisymmetry $\delta(k) + \delta(n - k) = [n]\tau(G)$	exact, public
22–29	coord change; formal log; Mazur–Tate; Mahler	no structure
30–36	Sage validation; 2-cocycle; canonical lift; Iwasawa	inert
37	Gowers U^2 anomaly	false positive (this paper)
40,40b	increment-shuffle and half-sequence controls	localizes to 21b
41	numerical verification of Theorem 1	confirmed
42	anomalous boundary (Smart)	12/12 recovery
43	clean-ordinary re-run + positive control	confirms §5
44–46	aggregation/MI estimators; alt. uniformizers; OEIS scan	post-submission, see repository
47	large-prime-order confirmatory curve ($p = 10477$)	confirms §5, §9
48	U^2 growth law (Prop. 1: $3/2$ mass, $\sqrt{n}$)	artifact signature

Table 3: Index of the exploration. Full scripts, fixed seeds, and JSON outputs are in the `ecdsa-lift` repository.

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